ORIGINAL ARTICLE

Diphtheria case detection and carrier study of close contacts: A step towards diphtheria eradication

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ABSTRACT

Introduction: Diphtheria is an infectious disease caused by gram-positive bacilli *C. diphtheriae* involving nasal, pharyngeal, tonsillar, or laryngeal mucus membranes. The mortality rate is as high as 20%, with India contributing almost 78% of the world incidence. **Aims and Objectives:** We report a fatal case of nasopharyngeal diphtheria with carrier study in close contacts. **Materials and Methods:** Seven years child presented with fever, throat pain, and earache for 3 days followed by neck swelling and noisy respiration. On examination, membrane was present in the throat, which was received for Albert and Gram staining and reported as positive for *C. diphtheria* like organisms followed by culture. The patient was treated with ADS and antibiotics, and intensively managed, but still succumbed to death. Follow-up was done for carriage of *C. diphtheriae* on the throat and nasopharyngeal swabs of siblings and close contacts. It was isolated in 3 of them. Samples were processed for Gram, Albert stain, and culture. Identification, antibiotic sensitivity, and toxigenicity were done. **Results and Discussion:** Four samples, one from the patient and three from contacts showed the presence of gram-positive slender bacilli with cuneiform arrangement, less cellular infiltrate on the Gram stain, and the presence of few metachromatic granules in the Albert stain. *C. diphtheriae* was grown on Potassium Tellurite agar. Antibiogram of all isolates was similar with resistance to Erythromycin and sensitivity to Penicillin. Isolates were confirmed by PCR and ToxA gene was detected. Contacts were treated with Penicillin and repeat swabs were negative. **Conclusion:** Present health statistics and this study suggests, fight against diphtheria in India is far from being over. It still lurks in some remote areas. It is a need to remain vigilant, keep tracing, and treating contacts to curtail down the rate of infection. In view of the resurgence, Government has given directives to replace TT with Td in UIP. Still, a lot needs to be done.

KEY WORDS: Carrier, contacts, diphtheria

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INTRODUCTION

With the advent of EPI in 1978 and UIP in 1985 most of the vaccine-preventable diseases have shown a decline but diphtheria is still endemic in our country. Recently 12 children died in 13 days in 2 Government Hospitals in Delhi. WHO case definition of diphtheria is an illness of the upper respiratory tract characterized by laryngitis or pharyngitis or tonsillitis and adherent membranes of tonsils, pharynx, and/or nose. Though diphtheria is on the verge of being eliminated in a few developed countries, 4530 cases were reported worldwide in 2015. Unfortunately 2365 (52.2%) of these were from India. The mortality rate is as high as 20%, with India contributing almost 78% of the world incidence.[1] WHO Vaccine-Preventable Diseases Monitoring System has also reported periodic resurgence in India.

Diphtheria is an infectious disease caused by gram-positive bacilli *Corynebacterium diphtheriae* involving nasal, pharyngeal, tonsillar, or laryngeal mucus membranes. The infection in respiratory diphtheria is transmitted via close contact with infectious material from respiratory secretions by droplet infection. Or it can spread through direct contact with skin lesions (Cutaneous diphtheria). Pathogenicity of diphtheria is toxin mediated. It elicits inflammatory response which leads to necrosis of the epithelium and exudate formation, forming mucosal ulcers, lined by tough leathery grayish white pseudomembrane coat.[2]

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Humans are the only known reservoir for *C. diphtheriae*. *Immunity*, either via natural infection or vaccine-induced does not prevent carriage. Hence asymptomatic carriers play an important role in disease transmission. Diagnosis of diphtheria remains primarily clinical supported by careful visualization of the pseudomembrane. Confirmatory laboratory diagnosis is by isolation of the diphtheria bacillus in tellurite-containing culture media. Toxigenicity tests detect the potent exotoxin, which is a phage-encoded protein. *In vivo* virulence testing in guinea pigs was replaced by Elek's test and now rapid toxin detection can be done by PCR.

MATERIALS AND METHODS

Case history 1

Seven years male child presented with fever, throat pain, and earache for 3 days followed by neck swelling ("Bull-neck" appearance) [Figure 1] and noisy respiration-(stridor). He had received 2 doses of inj. Pentavac. On examination he had a fever, the membrane was present in the throat, which was received for Gram staining [Figure 2] and Albert staining [Figure 3] and was reported as positive for *C. diptheria* like organisms followed by culture. The membrane was inoculated on blood and Potassium tellurite blood agar (0.04% PT). Fresh Loeffler's serum slope was not available at that juncture. Small black colonies [Figure 4] were obvious after overnight incubation followed by an increase in their size after 48 hours. The colonies were black and shiny with raised center. The colony smear showed gram-positive club-shaped bacteria, few with the cuneiform arrangement [Figure 5]. A biochemical study was done. The organisms were catalase positive and urease negative; glucose and maltose were fermented without showing gas formation in Hiss serum water, non-H_2S producer, nitrate reduction positive, showing diffused turbidity in liquid media. It was identified as *C. mitis*.

The antimicrobial susceptibility testing was performed by disk diffusion method based on CLSI M45. It was sensitive to Penicillin and resistant to Erythromycin. The strain was further confirmed as *C. diphtheriae* by real-time PCR, and toxigenicity testing was confirmed by the detection of the Tox A gene by real-time PCR.

Total WBC count was 14500/cu mm, with 70% polymorphs and Metamylocytes + bands = 12, followed by toxic changes in WBC on the next day. USG showed B/L Cervical lymphadenopathy. Initially, the child was treated with IV fluids and antibiotics. The next day he showed signs of respiratory distress and was immediately intubated and ventilated. Due to lack of access to the orogastric route as tube could not be passed due to membrane and inj. Crystalline Penicillin and Procaine Penicillin were not available, inj. Clindamycin was given. Diphtheria antitoxin 40,000 IU was given after the test dose. The patient was treated with ADS and antibiotics, and intensively managed, but still succumbed. The cause of death was grade IV tonsillitis with pneumonia and myocarditis, shock, with acute ARDS, that is, due to complications.

Case II

A 6-year-old female presented with chief complaints of cough since 8 days. The patient was apparently asymptomatic till the past 8 days, when she developed a cough without expectoration, non-pertussoid, and there was no post-tussive vomiting. No history of fever, sore throat, vomiting, loose stools, weakness, neck swelling. There was nothing significant in the previous history. Family history was suggestive with history of death of sibling due to Dipthteria. It was LSCS delivery in view of non-progressive labor birth weight not known; the child cried immediately after birth, it was full term having no history of NICU admission and no other significant past history. The child was immunized according to the National Immunization program till three and a half months of age.

DPT booster was given. On examination the child was conscious with HR: 100/min. RR: 24/min and had mild throat congestion with Grade 3 tonsillar hypertrophy. On systemic examination, there was nothing significant. Investigations were done. CBC reports revealed Hb-11.5 gm%, TLC-16900, Polymorph-39%, and Lymphocyte 50% Platelets-414000. A peripheral smear was done on the next day, which showed leucocytosis and adequate platelets. The nasopharyngeal swab was collected on the same day, which was Albert's stain negative.

However, the throat swab culture grew *C. diphtheriae*. Sensitivity was done which revealed that the organism was sensitive to Amoxicillin, Gentamicin, Ceftriaxone, Cephalexin, Ciprofloxacin, Penicillin G. Tetracycline, Vancomycin and resistant to Erythromycin. Tab Penicillin V 2.5 Lakh IU p/o BD for 14 days was advised.

The patient was followed up in pediatric OPD after 14 days and one month, respectively. The nasopharyngeal swab was collected after 14 days. It was negative for *C. diphtheriae* and the patient was clinically better.

Family screening importance was discussed in detail with the relatives and follow-up was done for carriage of *C. diphtheriae*

Figure 1: Patient with Diphtheria showing bull-neck appearance

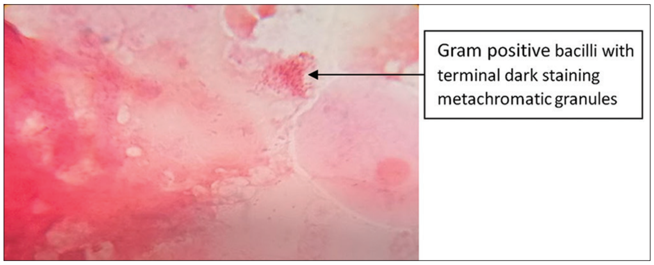

Figure 2: Gram staining of pseudomembrane

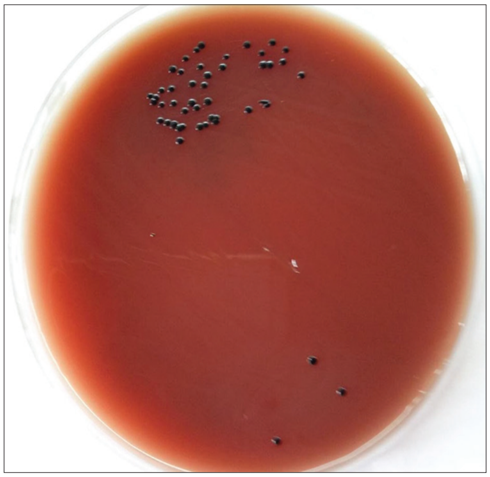

Figure 4: Black colonies of *C. diphtheriae* on Potassium tellurite agar

on the throat and nasopharyngeal swabs of siblings and close contacts Total of six children were screened. *C. diphtheriae* was isolated in 2 of them having age 6.5 and 7 years. Samples were processed for Gram, Albert stain, and culture. Identification, antibiotic sensitivity, and toxigenicity testing were done. All the positive contacts had incomplete immunization history. Booster doses were not given.

RESULTS AND DISCUSSION

Three samples: 1 from the patient and 2 from contacts showed the presence of gram-positive slender bacilli with cuneiform arrangement, less cellular infiltrate on Gram stain, and green bacilli with the presence of few metachromatic granules in Albert stain. *C. diphtheriae* was grown on Potassium tellurite agar. Antibiogram of all isolates was similar showing resistance to Erythromycin and sensitivity to Penicillin. AST was done by disc diffusion method on MH agar with 5% sheep blood agar according to CLSI. The strains were sensitive to Penicillin, Gentamicin, Amoxicillin, Ciprofloxacin, Cephalexin, Tetracyline, Ceftriaxone, Cloxacillin, Clindamycin, Vancomycin, Linezolid and resistant to Cotrimaxazole and Erythromycin. All the isolates showed similar antibiograms. Isolates were confirmed by PCR and Tox A gene was detected. Contacts were treated with oral Penicillin V for 14 days, repeat swabs were negative. All the isolates showed similar biotype and antibiogram indicating similarity between the strains. Malignant diphtheria is usually associated with extensive membrane pharyngitis along with massive swelling of the tonsils,

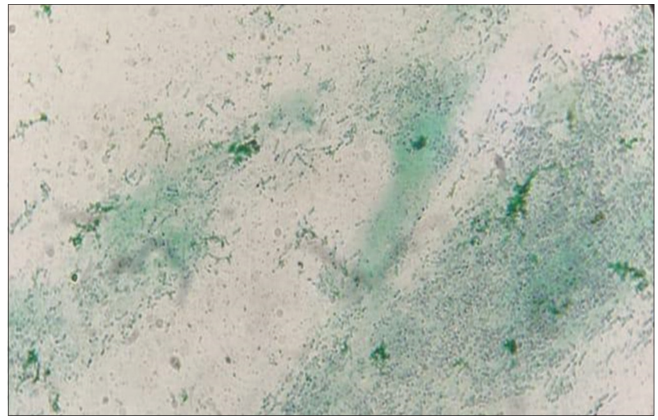

Figure 3: Albert staining of pseudomembrane

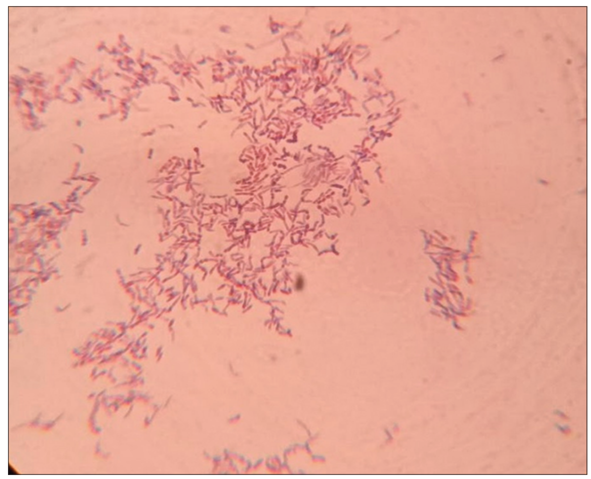

Figure 5: Gram staining of colony showing club-shaped bacilli

uvula, cervical lymph nodes, submandibular region, and anterior neck "Bull-neck" of toxic diphtheria. Respiratory stridor may ensue, leading to respiratory insufficiency and death as was seen in this case also. Diagnosis is usually clinical and treatment has to be started immediately after sample collection. One must not wait for lab confirmation. Once confirmation is done by culture, the importance lies in tracking and treating the carriers. The mainstay of treatment of diphtheria is antitoxins, which work by neutralizing the free toxins. A single dose of 20,000–1,00,000 units should be administered as soon as a clinical diagnosis is made. Antibiotic abort toxin production, treat localized infection, and prevent transmission to susceptible contacts. The antibiotics of choice are Penicillin or Erythromycin for 14 days after which two negative cultures from the nose and throat should be obtained.

The Vijayapura district of North Karnataka has become the hotbed of diphtheria constantly showing a resurgence in the region.[3] Contacts were also from Athani Taluka of Belgavi District, which is adjoining the Vijayapura district.

Though there are multiple causes of the resurgence of diphtheria, the shift in the age group affecting it is mainly attributed to a reduction in vaccination coverage during childhood, creating pockets of low immunization, waning adult immunity, large-scale population movements, disruptions in health service and inadequate supplies of vaccines and antitoxin. Toxigenic *C. diphtheria* continues to circulate, leading to the reemergence of the disease.[4] Elimination of *diphtheria* is possible as a human is the only reservoir and safe effective and affordable vaccine is available. As the outbreaks occur seasonally, transmission interruption is possible and contacts were also from Athani Taluka of Belgavi District, which is adjoining the Vijayapura district.

Symptoms of severe disease are mediated by an exotoxin produced by strains of *C. diphtheriae* which are infected by lysogenic bacteriophage possessing tox gene.[4] Confirmation of diphtheria is dependent on the isolation of *C. diphtheriae* and its toxigenicity testing by Elek's test. Elek's test detects an immunoprecipitation reaction between diphtheria toxin and antitoxin proving the toxigenicity. It requires 24 hours of time after the isolation of the organism. However, less number of laboratories carry out this test as there is a shortage of DAT (*Diptheria* antitoxin). Hence, detection of the toxin gene of *C. diphtheriae,* which is the major virulence factor by real-time PCR is now commonly considered as a rapid diagnostic method, compared to the standard method.

RTPCR has been proved to be a more sensitive method increasing the rate of positive detection.[4] It can also detect tox gene directly from clinically relevant samples and the turnaround test time is only 2-4 hours, while gold standard conventional culture and Elek's gel test will require a minimum of 48–72 hours.

RTPCR complements diagnosis by culture, as it can provide rapid and presumptive diagnosis directly from clinical specimens especially while responding to outbreak scenarios with limited culture facilities. The use of real-time PCR for confirmation of *C. diphtheriae* as the main cause of diphtheria and the diphtheria toxin gene as the major virulence factor directly from clinical specimens can be considered whenever possible and available. The main purpose is rapid and prompt treatment and infection control.[5]

Take home message

Preventing the preventable through effective surveillance is the key in reducing the number of diphtheria cases. A high level of clinical suspicion towards diphtheria should be maintained and early treatment should be initiated. Try, detect, and treat carriers and close contacts.[2] Mechanisms have been strengthened for case investigation, reporting, and data management.[3] Laboratory confirmation of clinically suspected diphtheria cases is crucial for accurately classifying cases and understanding the epidemiology and circulation of strains. Diphtheria being an emerging pathogen of increasing significance worldwide, it is imperative to take appropriate measures to control the disease and step further toward diphtheria eradication. The present health statistics and our study suggests, fight against diphtheria in India is far from being over.[6] It still lurks in some remote areas. As humans are the only reservoirs, and there is the availability of safe, effective, and efficient vaccines, it is possible to eradicate diphtheria. Due to seasonal outbreaks, the transmission can also be interrupted.[7] It is a need to remain vigilant and keep tracing and treating close contacts with their effective management to curtail down the rate of infection. It is a necessity to strengthen laboratory networks for genotyping and toxigenicity testing. Robust and effective surveillance for monitoring disease trends, clinical presentation, mutations leading to emerging strains and drug resistance is a way forward.[8] The role of microbiology goes beyond Petri dishes and molecular biology. Microbiology with the Department of Community Medicine can work together to fight the battle. At an individual level, each one of us can play an important role in the journey towards the elimination of diphtheria in India. In view of the resurgence, Government has given directives to replace TT with Td in UIP. Still, a lot needs to be done.

Acknowledgement

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Declaration of patient consent

We certify that we have obtained all appropriate patient consent forms. In the form, the patient (s) has/have given his/her/their consent for his/her/their images and other clinical information to be reported in the journal. The patients understand that their names and initials will not be published and due efforts will be made to conceal their identity, but anonymity cannot be guaranteed.

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Nil.

Conflict of interest

There are no conflicts of interest.

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